

Tushar Alase

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EDUCATION

Purdue University – West Lafayette, IN

Aug 2026 – May 2028 (Expected)

Master of Science in Electrical and Computer Engineering

Savitribai Phule Pune University – Pune, India

Oct 2020 – Apr 2023

Bachelor of Engineering in Electronics and Telecommunication Engineering Honours: Internet of Things – GPA: 9.11/10.00

EXPERIENCE

Embedded & Edge Systems Consultant (Freelance)

Mar 2026 – Jul 2026

AI Unika

Remote

- Advised teams on platform selection and system integration across 4 edge-AI and IoT products using Raspberry Pi, Jetson, cameras, and sensors.
- Helped select LiDAR and acquire camera, depth, and GPS data for real-time 3D mapping of roadside structures.

Senior Embedded System Developer

Mar 2024 – Feb 2026

Cyamsys Technologies

Noida, India

- Adapted TI SDK firmware for a BLDC motor driver, refining speed and direction control, feedback, HMI integration, and responses to voltage, current, and thermal faults; implemented Modbus communication with external modules.
- Developed firmware for BLDC- and DC-motor sliding-door controllers using TI drivers, integrating Hall-sensor feedback, configurable acceleration, end-of-travel deceleration, and travel calibration for 300–400 kg glass doors.
- Implemented obstruction-triggered reversal and a seven-segment interface for speed, operating modes, test mode, slowdown distance, and detection sensitivity; tuned Hall-feedback filtering to distinguish obstructions from irregular door movement and wear.
- Architected ESP32 FOTA with version checks and separate firmware partitions, integrating HTTP over Wi-Fi and FTP through an EC200U 4G module; supported 150 devices with 95% first-attempt update success and rollback to the previous working image for the remaining 5%.
- Developed MSP430 firmware-update workflows, first transferring images from a Python tool over USB for testing, then adapting the update path to FTP delivery over 4G.
- Automated firmware testing with Python, reducing validation time from 3 hours to 90 minutes across 6 projects.

SELECTED PROJECTS

Road-Anomaly Detection and V2V Alert Prototype

- Developed a YOLOv8-based pothole-detection pipeline on Jetson Nano for camera-based driver alerts.
- Recorded detected potholes with GPS coordinates on a server for location-based hazard tracking.
- Broadcast hazard data via a 2.4 GHz nRF24L01+ transceiver for vehicle-to-vehicle alerts.

IIoT Data Logger & Remote Device Management System

- Developed an ESP32-based industrial data logger integrating Modbus device data with remote telemetry and diagnostics.
- Architected BLE communication with a mobile app for live monitoring and device configuration.
- Implemented configurable reporting frequency and phone-to-device time synchronization with the on-board RTC.

AutOmelette Articulated Cooking Robot

- Programmed coordinated six-servo movements for an automated omelette-cooking robotic arm.
- Integrated interchangeable oil-brush, ladle, and spatula operations into a timed cooking and serving sequence.
- Integrated ESP8266 remote-start functionality through Google Assistant, IFTTT, and Adafruit IO, including a geofence-triggered start.

TECHNICAL SKILLS

Programming: C, C++, Embedded C, Python

Embedded & Edge Platforms: ESP32/ESP-IDF, STM32, MSP430/MSPM0, ATmega4809, ARM Cortex-M, Raspberry Pi, NVIDIA Jetson, FreeRTOS, Embedded Linux

Communication & IoT: UART, SPI, I2C, RS-485, Modbus, BLE, Wi-Fi, LoRa, GSM/4G, MQTT, HTTP/FTP, telemetry, remote diagnostics

Tools & Hardware: Git/GitHub, Code Composer Studio, STM32CubeIDE, PlatformIO, KiCad, JTAG, logic analyzers, sensor integration, board bring-up

SELECTED AWARDS

Technology Infusion Grand Challenge Asia Funding, La Trobe University, 2022 | **Audience Choice Award**, Circuit Digest, 2021 | **Winner**, Abhiyantrix, 2023